

AAEC 5301: Applied Mathematical Economics

Week 1, Lecture 2

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Where we left off

On Tuesday:

- Mathematical economics is an approach, not a field. Its virtue is that it forces assumptions into the open.
- A model is a system of equations: definitional, behavioral, conditional.
- Endogenous versus exogenous is a modeling decision, not a property of a variable.
- Sets, set operations, and the difference between verifying and proving.

Today we finish Chapter 2. The destination is the sentence “let $Q_d = D(P)$ be a demand function” and understanding everything that sentence is quietly claiming.

Model-based versus design-based economics

Two ways to do applied microeconomics. Both mainstream, both with Nobel prizes.

Model-based	Design-based
Write down how agents choose: what they want, know, constrains them	Find variation in the world that is as good as random
Estimate what lies behind agents' behavior: costs, preferences, demand	Estimate the effect of a change that actually happened
Then ask what <i>would</i> happen in a 'counterfactual' world: a merger, a new fee	Credible for that change, in that place, at that time
Only as good as its assumptions	Silent on anything the variation does not happen to ask

Design-based: Card, Angrist, Imbens (2021). **Model-based:** Myerson and Maskin on mechanism design (2007), Roth and Shapley on market design (2012), Tirole on market power (2014), Milgrom and Wilson on auctions (2020).

Two papers, two questions

Design-based

Duflo and Pande (2007), *QJE*

- Estimates what the dams that *were* built did
- Needs the policy to have happened (cannot be a counterfactual)

JOURNAL ARTICLE

Dams*

Esther Duflo, Rohini Pande

The Quarterly Journal of Economics, Volume 122, Issue 2, May 2007, Pages 601–646,

<https://doi.org/10.1162/qjec.122.2.601>

Published: 01 May 2007

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Abstract

This paper studies the productivity and distributional effects of large irrigation dams in India. Our instrumental variable estimates exploit the fact that river gradient affects a district's suitability for dams. In districts located downstream from a dam, agricultural production increases, and vulnerability to rainfall shocks declines. In contrast, agricultural production shows an insignificant increase in the district where the dam is located but its volatility increases. Rural poverty declines in downstream districts but increases in the district where the dam is built, suggesting that neither markets nor state institutions have alleviated the adverse distributional impacts of dam construction.

Model-based

Roth, Sönmez and Ünver (2004), *QJE*

- Paying for organs is illegal, so there are no prices to observe
- Models patients, donors, and feasible organ exchange
- Describes an institution that did not exist; then it was built

Kidney Exchange*

Alvin E. Roth, Tayfun Sönmez, M. Utku Ünver

The Quarterly Journal of Economics, Volume 119, Issue 2, May 2004, Pages 457–488,

<https://doi.org/10.1162/0033535041382457>

Published: 01 May 2004

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Abstract

Most transplanted kidneys are from cadavers, but there are also many transplants from live donors. Recently, there have started to be kidney exchanges involving two donor-patient pairs such that each donor cannot give a kidney to the intended recipient because of immunological incompatibility, but each patient can receive a kidney from the other donor. Exchanges are also made in which a donor-patient pair makes a donation to someone waiting for a cadaver kidney, in return for the patient in the pair receiving high priority for a compatible cadaver kidney when one becomes available. There are stringent legal/ethical constraints on how exchanges can be conducted. We explore how larger scale exchanges of these kinds can be arranged efficiently and incentive compatibly, within existing constraints. The problem resembles some of the "housing" problems studied in the mechanism design literature for indivisible goods, with the novel feature that while live donor kidneys can be assigned simultaneously, cadaver kidneys cannot. In addition to studying the theoretical properties of the proposed kidney exchange, we present simulation results suggesting that the welfare gains from larger scale exchange would be substantial, both in increased number of feasible live donation transplants, and in improved match quality of transplanted kidneys.

Why this matters in the first week of a math course

- The two are **complements**, not rivals. An economic model needs credible variation to be identified. A reduced-form estimate needs a model before it can say anything about a policy that has not happened yet.
- Most good empirical work now uses both: clean variation to pin down a few parameters, a model to carry them in a counterfactual setting.
- But only one of the two requires you to **write the model down**, and that is what this semester is for. Optimization, constraints, comparative statics (tools for model-based economics). More such tools in micro-theory, game theory, industrial organization economics.

You will get the design-based tools in econometrics. **JMP advice:** marry the two methods.

More info: <https://www.aeaweb.org/articles?id=10.1257/jep.24.2.69> and
<https://www.nber.org/papers/w15794>

Today

1. §2.4 Ordered pairs, Cartesian products, relations, and functions
2. §2.5 Types of function, and a digression on exponents
3. §2.6 Functions of two or more independent variables
4. §2.7 Levels of generality
5. Application: writing a demand curve as a function

First, the one from Tuesday. Exercise 2.3, problem 9: how can $\emptyset$ be both a subset of U and the complement of U ?

The null-set paradox

The apparent contradiction: “complement of U ” sounds like being *outside* U , while “subset of U ” sounds like being *inside* it.

The null-set paradox

The apparent contradiction: “complement of U ” sounds like being *outside* U , while “subset of U ” sounds like being *inside* it.

There is no contradiction, because the two statements are about different things.

- $\tilde{U} = \emptyset$ answers “what is outside U ?” Nothing is, because U is everything we are considering.
- $\emptyset \subset U$ answers “is $\emptyset$ inside U ?” Yes. To fail, $\emptyset$ would need an element that U lacks, and it has no elements at all.

The word “in” is doing two different jobs. That is exactly why we distinguish $\in$ from $\subset$.

Section 2.4

Relations and Functions

Ordered pairs

In writing the set $\{a, b\}$ we do not care about order: $\{a, b\} = \{b, a\}$. That is an **unordered** pair.

When order carries meaning we write the **ordered pair** (a, b) , with

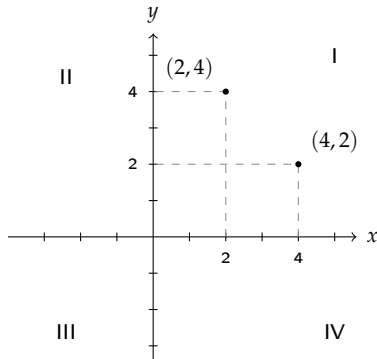
$$(a, b) \neq (b, a) \quad \text{unless } a = b.$$

Parentheses, not braces.

The same idea extends to ordered triples, quadruples, and so on.

Example. Record each student's age and weight as (a, w) . Then $(19, 127)$ and $(127, 19)$ mean very different things, and the second would not fit any student anywhere.

Ordered pairs are points in the plane



The xy plane is an infinite set of points. Each point *is* an ordered pair: an x value first, a y value second.

The point $(2, 4)$ is not the point $(4, 2)$. Ordering matters here, which is why the plane is built from ordered pairs and not from sets.

Cartesian products

Given two sets, form every ordered pair with the first element from the first set and the second element from the second. With $x = \{1, 2\}$ and $y = \{3, 4\}$:

$$x \times y = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

read “ x cross y .” In general,

$$x \times y = \{(a, b) \mid a \in x \text{ and } b \in y\}.$$

Note what changed. x and y are sets of numbers. Their Cartesian product is a set of *ordered pairs*. The operation does not just combine elements; it changes what kind of object we are holding.

From $x \times y$ to $\mathbb{R}^n$

Now let both x and y be all of the real numbers:

$$x \times y = \{(a, b) \mid a \in \mathbb{R} \text{ and } b \in \mathbb{R}\} = \mathbb{R} \times \mathbb{R} = \mathbb{R}^2.$$

Every ordered pair corresponds to a unique point in the plane, and every point to a unique ordered pair. That double uniqueness is a **one-to-one correspondence**, and it is why the notation $x \times y$ echoes the crossing of the two axes.

Extending:

$$x \times y \times z = \{(a, b, c) \mid a \in x, b \in y, c \in z\}, \quad \mathbb{R} \times \mathbb{R} \times \mathbb{R} = \mathbb{R}^3,$$

and in general $\mathbb{R}^n$. Everything in this course will be in $\mathbb{R}^n$ for $n \in \mathbb{N}$.

Concept check 1

Concept check

Let $S_1 = \{3, 6, 9\}$, $S_2 = \{a, b\}$, $S_3 = \{m, n\}$.

1. How many elements does $S_1 \times S_2$ have?
2. Write out $S_2 \times S_3$.
3. Is $S_2 \times S_3 = S_3 \times S_2$?

Concept check 1

Concept check

Let $S_1 = \{3, 6, 9\}$, $S_2 = \{a, b\}$, $S_3 = \{m, n\}$.

1. How many elements does $S_1 \times S_2$ have?
2. Write out $S_2 \times S_3$.
3. Is $S_2 \times S_3 = S_3 \times S_2$?

Answer. 1: $3 \times 2 = 6$. 2: $\{(a, m), (a, n), (b, m), (b, n)\}$. 3: No.

$S_3 \times S_2 = \{(m, a), (m, b), (n, a), (n, b)\}$, and $(a, m) \neq (m, a)$.

Follow-up: under what condition *would* the two products be equal? If two sets are equal.

Relations

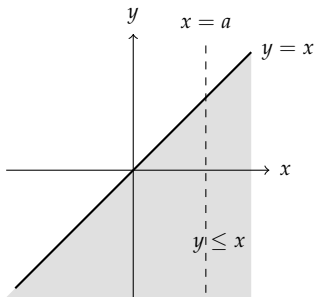
Any subset of $\mathbb{R} \times \mathbb{R}$ is a **relation** between y and x . Pick some collection of ordered pairs; that collection is a relation.

Example. $\{(x, y) \mid y = 2x\}$ contains $(1, 2)$, $(0, 0)$, $(-1, -2)$, and so on.

Example. $\{(x, y) \mid y \leq x\}$ contains $(1, 0)$, $(1, 1)$, and $(1, -4)$.

Given an x value, a relation specifies **one or more** y values. In the second example, $x = 1$ is compatible with $y = 0$, $y = 1$, and $y = -4$ all at once.

A function is a relation with a uniqueness property



A relation, not a function

As a **special case**, a relation may assign exactly one y to each x . Then y is a **function** of x , written $y = f(x)$.

Graphically: no vertical line meets the graph more than once. The dashed line at $x = a$ crosses the shaded region infinitely often, so $y \leq x$ fails.

Every function is a relation. Not every relation is a function.

Note: $f(x)$ does not mean f times x .

Uniqueness is required in one direction only

The definition requires a unique y for each x . It does **not** require a unique x for each y .

Two different x values may map to the same y . The function $y = x^2$ sends both $x = 2$ and $x = -2$ to $y = 4$, and it is a perfectly good function.

Terminology note. What we call a function was once called a *single-valued* function. What was once called a *multivalued* function is now called a relation, or a **correspondence**. You will meet correspondences again in game theory, where a best-response correspondence can return a set.

Functions as mappings

A function is also called a **mapping** or a **transformation**. Both words suggest the action of associating one thing with another:

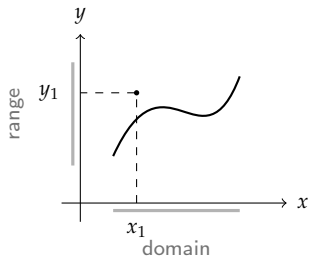
$$f : x \rightarrow y$$

where the arrow indicates mapping and f names the rule.

Since f is a *particular* rule, a second function in the same model needs a different name: g , F , G , ϕ , ψ , Φ , Ψ .

So if $y = f(x)$, write $z = g(x)$ for the other one — not $z = f(x)$. It is also acceptable to drop the letters entirely and write $y = y(x)$ and $z = z(x)$.

Argument, value, domain, range



- x is the **argument**, or independent variable.
- y is the **value**, or dependent variable.
- The **domain** is the set of all permissible x values.
- The y that a given x maps into is that x 's **image**.
- The **range** is the set of all images.

Domain belongs to x ; range belongs to y .

Domains in economics

In economic models, behavioral equations enter as functions. Most economic variables are restricted to **nonnegative** real numbers, so their domains are too.

That is why economic diagrams are usually drawn in the first quadrant only.

We will not specify a domain for every function in every model. When none is given, understand the domain and range to include only the numbers for which the function makes **economic** sense.

“Nonnegative” rather than “positive”: zero output, zero consumption, and zero input use are all permissible.

Example: a cost function with a capacity limit

A firm's daily total cost is $C = 150 + 7Q$, and it can produce at most 100 units a day.

$$\text{Domain} = \{Q \mid 0 \leq Q \leq 100\}$$

The function is a straight line, so C is smallest at $Q = 0$ and largest at $Q = 100$:

$$C(0) = 150, \quad C(100) = 850, \quad \text{Range} = \{C \mid 150 \leq C \leq 850\}$$

But be careful. That reasoning worked because the function is monotonic. In general the extreme values of the range need **not** occur at the endpoints of the domain.

Consider $y = -x^2 + 4x$ on $0 \leq x \leq 4$. Both endpoints give $y = 0$; the maximum is at $x = 2$, in the interior. Finding such interior maxima is Chapters 9 through 13.

Concept check 2

Concept check

1. Drawn in the plane, which of these is the graph of a function: a circle, a triangle, a downward-sloping straight line?
2. If $y = 5 + 3x$ with domain $\{x \mid 1 \leq x \leq 9\}$, what is the range?
3. Economists write $C = f(Q)$. Must each level of output determine a unique cost? Must each cost figure be associated with a unique output?

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Answer. 1: Only the line. A circle and a triangle both fail the vertical-line test. 2: $\{y \mid 8 \leq y \leq 32\}$. 3: Yes to the first, no to the second — uniqueness runs from argument to value only.

Section 2.5

Types of Function

Constant and polynomial functions

$y = f(x)$ says a mapping exists but does not say what the rule is. Now we name some rules.

A **constant function** has a range with one element: $y = f(x) = 7$, whatever x is. A horizontal line. In national-income models, exogenous investment $I = I_0$ is exactly this.

A **polynomial function** (“multiterm”) of one variable has the general form

$$y = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

where every term has a coefficient and a nonnegative-integer power of x .

Why subscripted coefficients $a_0, a_1, \dots$ rather than a, b, c ? Economy of alphabet, and the subscript tells you where the coefficient sits: a_2 always multiplies x^2 .

Subclasses by degree

$n = 0:$ $y = a_0$ constant function

$n = 1:$ $y = a_0 + a_1x$ linear function

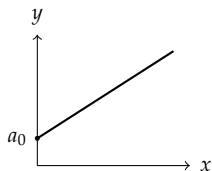
$n = 2:$ $y = a_0 + a_1x + a_2x^2$ quadratic function

$n = 3:$ $y = a_0 + a_1x + a_2x^2 + a_3x^3$ cubic function

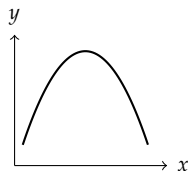
The superscripts are **exponents**; the highest one, n , is the **degree** of the polynomial. A quadratic is a second-degree polynomial, a cubic a third-degree one.

The last coefficient is always assumed nonzero, otherwise the function collapses to a lower degree.

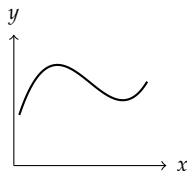
What they look like



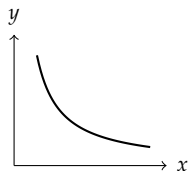
linear



quadratic, $a_2 < 0$



cubic



$xy = a$

For the linear case, a_0 is the vertical intercept and a_1 is the slope: a one-unit increase in x raises y by a_1 . The quadratic is a parabola, a hill when $a_2 < 0$ and a valley when $a_2 > 0$. The cubic generally shows two wiggles.

Rational functions

A **rational function** expresses y as a ratio of two polynomials:

$$y = \frac{x - 1}{x^2 + 2x + 4}$$

By this definition every polynomial is itself rational: write it as a ratio to 1, and 1 is a constant function.

One special case matters a great deal in economics:

$$y = \frac{a}{x} \quad \text{equivalently} \quad xy = a$$

which plots as a **rectangular hyperbola**. The curve approaches both axes asymptotically and never touches them.

Why the rectangular hyperbola shows up twice

A unit-elastic demand curve.

With P and Q on the axes, $PQ = a$ says total expenditure is the same constant at every price. Spending does not respond to price at all, which is precisely unitary elasticity at every point on the curve.

The average fixed cost curve.

$AFC \times Q = TFC$, and total fixed cost is a constant by definition. So AFC against Q *must* be a rectangular hyperbola. Not an assumption — an identity.

Same curve, two entirely different economic stories. This is worth noticing: functional forms are portable, and recognizing one you have seen before is most of what makes a model readable.

Nonalgebraic functions

A function built from polynomials and roots of polynomials is **algebraic**. Everything so far has been.

Two important types are not:

- **Exponential** functions, $y = b^x$, where the variable is in the exponent.
- **Logarithmic** functions, $y = \log_b x$.

These are **transcendental** functions. We defer them to Chapter 10 (Week 9), where they earn their own treatment through growth, compounding, and discounting.

Trigonometric functions are also transcendental. Chiang treats them in Chapter 16.

Concept check 3

Concept check

Classify each: constant, linear, quadratic, cubic, rational (non-polynomial), or nonalgebraic.

1. $y = 4$

2. $y = 3 - 2x + x^2$

3. $y = 100/x$

4. $y = 2^x$

5. $y = \sqrt{x}$

Concept check 3

Concept check

Classify each: constant, linear, quadratic, cubic, rational (non-polynomial), or nonalgebraic.

1. $y = 4$

2. $y = 3 - 2x + x^2$

3. $y = 100/x$

4. $y = 2^x$

5. $y = \sqrt{x}$

Answer. 1 constant. 2 quadratic. 3 rational, non-polynomial. 4 nonalgebraic — the variable is in the exponent. 5 algebraic but not a polynomial: it is a root of a polynomial, and $\sqrt{x} = x^{1/2}$ is not a nonnegative-integer power.

A digression on exponents

For a positive integer n , define $x^n = \underbrace{x \times x \times \cdots \times x}_{n \text{ terms}}$, with $x^1 = x$.

Rule I. $x^m \times x^n = x^{m+n}$ e.g. $x^3 \times x^4 = x^7$

Proof. The left side is m terms of x multiplied by n terms of x , which is $m + n$ terms of x .

Rule II. $\frac{x^m}{x^n} = x^{m-n} \quad (x \neq 0)$

Proof. The n terms in the denominator cancel n of the m terms in the numerator, leaving $m - n$.

$x = 0$ is excluded because x^m / x^n would be division by zero.

Rules III and IV fall out of Rule II

What if $m < n$? Take $m = 2$, $n = 5$:

$$\frac{x^2}{x^5} = \frac{x \cdot x}{x \cdot x \cdot x \cdot x \cdot x} = \frac{1}{x^3} \quad \text{while Rule II says it is } x^{-3}.$$

Rule III. $x^{-n} = \frac{1}{x^n}$ ($x \neq 0$). A negative power means the reciprocal of the positive power.

And what if $m = n$? Rule II gives x^0 , while direct cancellation gives 1:

Rule IV. $x^0 = 1$ ($x \neq 0$). The expression 0^0 is undefined.

Notice that neither rule was invented. Both were forced on us by requiring Rule II to keep working outside the case it was proved for. That is how mathematics usually extends a definition.

Fractional powers, and the last two rules

Polynomials need only nonnegative integer powers. Exponential functions do not, so we need to interpret something like $x^{1/2}$. By Rule I,

$$x^{1/2} \times x^{1/2} = x^{1/2+1/2} = x.$$

Whatever $x^{1/2}$ is, multiplying it by itself gives x . So it is the square root.

Rule V. $x^{1/n} = \sqrt[n]{x}$

Rule VI. $(x^m)^n = x^{mn}$

Rule VII. $x^m \times y^m = (xy)^m$

A callback. Look at how Rule I was proved: no value was assigned to x , m , or n . That is what makes it a proof rather than a verification — exactly the distinction from Tuesday.

Concept check 4

Concept check

Condense or evaluate.

1. $x^4 \times x^{15}$

2. x^a / x^{-a}

3. $(x^{1/2} \times x^{1/3})^6$

4. Show that $x^{-3/2} = 1/\sqrt{x^3}$, naming the rule at each step.

Concept check 4

Concept check

Condense or evaluate.

1. $x^4 \times x^{15}$

2. x^a / x^{-a}

3. $(x^{1/2} \times x^{1/3})^6$

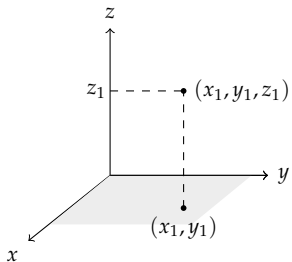
4. Show that $x^{-3/2} = 1/\sqrt{x^3}$, naming the rule at each step.

Answer. 1: x^{19} (Rule I). 2: $x^{a-(-a)} = x^{2a}$ (Rule II). 3: $(x^{5/6})^6 = x^5$ (Rules I then VI). 4: $x^{-3/2} = 1/x^{3/2}$ by III; $x^{3/2} = (x^3)^{1/2}$ by VI; $(x^3)^{1/2} = \sqrt{x^3}$ by V.

Section 2.6

Functions of Two or More Independent Variables

More than one argument



Given $z = g(x, y)$, a *pair* of values (x, y) determines a unique z .

So the domain is no longer a set of numbers. It is a set of **ordered pairs** or a subset of the xy plane, $\mathbb{R}^2$.

Erect a z axis and the value of the function at a point in the domain is the height of a vertical line planted there. The locus of such heights is a **surface**.

Where $y = f(x)$ is a set of ordered pairs, $z = g(x, y)$ is a set of ordered triples.

The obvious economic application

If output is determined by capital K and labor L , write the **production function**

$$Q = Q(K, L).$$

This is a function of two variables in exactly the above sense: a pair of input levels determines a unique output level.

Notice we have written it in *general form*. We have not said whether it is Cobb–Douglas, CES, or something else. That is deliberate, and §2.7 is about why.

Week 6: marginal products are the partial derivatives of this function. Week 12: Cobb–Douglas and CES get their own treatment.

Beyond three dimensions

With $y = h(u, v, w)$ we map a point in three-dimensional space into a point on a line.

A consumer's utility as a function of three commodities is exactly this: a mapping from a three-dimensional commodity space into a one-dimensional utility space.

But we cannot draw it. Picturing ordered quadruples would take four dimensions, and we do not have four.

So we borrow the language anyway. We call (u_1, v_1, w_1, y_1) a **point** in four-dimensional space, and the locus of such points a **hypersurface**. Both terms carry over to the general n -dimensional case.

This is the Chapter 1 argument arriving. Geometry stopped at two goods. The notation does not stop at all.

Types, and a note on notation

A function of several variables is **linear** if every variable appears to the first power only:

$$y = a_1x_1 + a_2x_2 + \cdots + a_nx_n$$

It is **quadratic** if it involves first and second powers, with the sum of exponents in any single term no greater than 2. So x_1x_2 is a quadratic term; so is x_1^2 .

Why $x_1, x_2, \dots, x_n$ rather than x, y, u, v ? The same two reasons as for subscripted coefficients: economy of alphabet, and an easier accounting of how many variables are in play. You will be grateful for this by Week 10.

Concept check 5

Concept check

1. Is $z = 3x_1 + 4x_2 - x_1x_2$ linear, quadratic, or neither?
2. For $Q = Q(K, L)$, what kind of object is the domain?
3. A utility function over four goods: can it be graphed? What do we call its “graph”?

Concept check 5

Concept check

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2. For $Q = Q(K, L)$, what kind of object is the domain?
3. A utility function over four goods: can it be graphed? What do we call its “graph”?

Answer. 1: Quadratic. The term x_1x_2 has exponents summing to 2, which disqualifies it from being linear but is allowed in a quadratic. 2: A set of ordered pairs (K, L) , in practice the nonnegative quadrant. 3: No — it would need five dimensions. We call it a hypersurface.

Section 2.7

Levels of Generality

Three levels

1. Numerical. $y = 7$, $y = 6x + 4$, $y = x^2 - 3x + 1$

Numerical coefficients, and the type is pinned down. Each gives one well-defined curve and numerical solutions.

2. Parametric. $y = a$, $y = a + bx$, $y = a + bx + cx^2$

Each represents not one curve but a **whole family** of curves.

3. General function. $y = f(x)$, $z = g(x, y)$

Not restricted to being linear, quadratic, exponential, or anything else. All of those are subsumed.

What each level costs and buys

	Buys you	Costs you
Numerical	A number you can take to a policymaker	Change one coefficient and you redo the whole derivation
Parametric	One derivation covers a family; assign values afterward	The functional form is still assumed
General	Conclusions that hold for any function of that shape	Answers are qualitative in terms of signs, but not magnitudes

There is no “best” level. The right choice depends on whether you need a number or a theorem. Most of this course lives at level 2 and 3; your econometrics course drags you back to level 1.

General functions still need restrictions

Working at level 3 has a catch. To get economically meaningful results, we usually must impose **qualitative restrictions** on the general functions in the model. For example:

- a demand function must have negatively sloped graph (**exception: veblen goods**);
- a consumption function $C = C(Y)$ must have a positive slope less than 1.

Notice what these restrictions are: they are not functional forms. They are statements about *signs*. That is enough to sign a comparative-static result without ever choosing a form, which is the whole method of Chapter 8, and the Week 7 application on your syllabus.

Application: writing a demand curve as a function

Same demand relationship, three levels:

$Q_d = 100 - 2P$	numerical
$Q_d = a - bP, \quad a, b > 0$	parametric
$Q_d = D(P), \quad D'(P) < 0$	general

- Writing $Q_d = D(P)$ **asserts** that each price determines exactly one quantity demanded. That is a modeling assumption, not an observed fact.
- Domain: $P \geq 0$. Range: nonnegative quantities. First quadrant only.
- $b > 0$ and $D'(P) < 0$ say the same thing.

And where the endogenous/exogenous line reappears

Write the demand function more fully:

$$Q_d = D(P; Y_0, P_0^s)$$

with income Y_0 and the price of a substitute P_0^s carrying the exogenous subscript.

- Change P : you **move along** the curve. P is endogenous, and the model determines it.
- Change Y_0 : the whole curve **shifts**. Y_0 is exogenous, handed to the model from outside.

Movement along versus shift of a curve is about endogenous/exogenous distinction from the last lecture, drawn on a diagram.

Chapter 2 in one slide

The structure of a mathematical economic model:

- A system of equations: definitional, behavioral, or equilibrium conditions.
- The behavioral equations are usually functions: linear or nonlinear, numerical or parametric, one independent variable or many.
- It is through these that the model's assumptions are given mathematical expression.

And the procedure, in order:

1. Select the variables, endogenous and exogenous.
2. Translate the assumptions — behavioral, technological, institutional, legal — into equations.
3. *Only then* derive conclusions, and give them economic interpretations.

Inequalities can also enter a model. We set them aside until Chapter 13.

For next week

Read: Chiang and Wainwright, Chapter 3 — Equilibrium Analysis in Economics.

We will cover: the meaning of equilibrium; partial market equilibrium in linear and nonlinear models; the n -commodity case; equilibrium in national-income analysis.

Note the connection. Today's market model had three equations and three endogenous variables but we never solved it. Next week we solve it, and the conditional equation $Q_d = Q_s$ is what makes that possible.

Also: Problem Set 1 is due at the end of Week 2 — models, sets, functions, and equilibrium. There will be a short quiz at the start of Tuesday.

Questions?